

Lesson 8: One-Sample t-Intervals and Population Proportions

BSTA 551: Statistical Inference

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Lesson 8: The Realistic Case

Review: Where We Left Off

Lesson 7: Confidence Intervals (Introduction)

- CI definition and correct interpretation
- Pivotal quantities for constructing CIs
- Z-intervals when σ is known: $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

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The Problem: In practice, we **rarely know** the population standard deviation σ .

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The Problem: In practice, we **rarely know** the population standard deviation σ .

Today's Goals:

1. Construct t-intervals when σ is unknown
2. Check normality assumptions
3. Build confidence intervals for population proportions
4. Compare Wald, Score, and Agresti-Coull intervals

CI for Mean: σ Unknown

The Realistic Case

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Natural idea: Replace σ with the sample standard deviation s .

But what happens to our pivotal quantity?

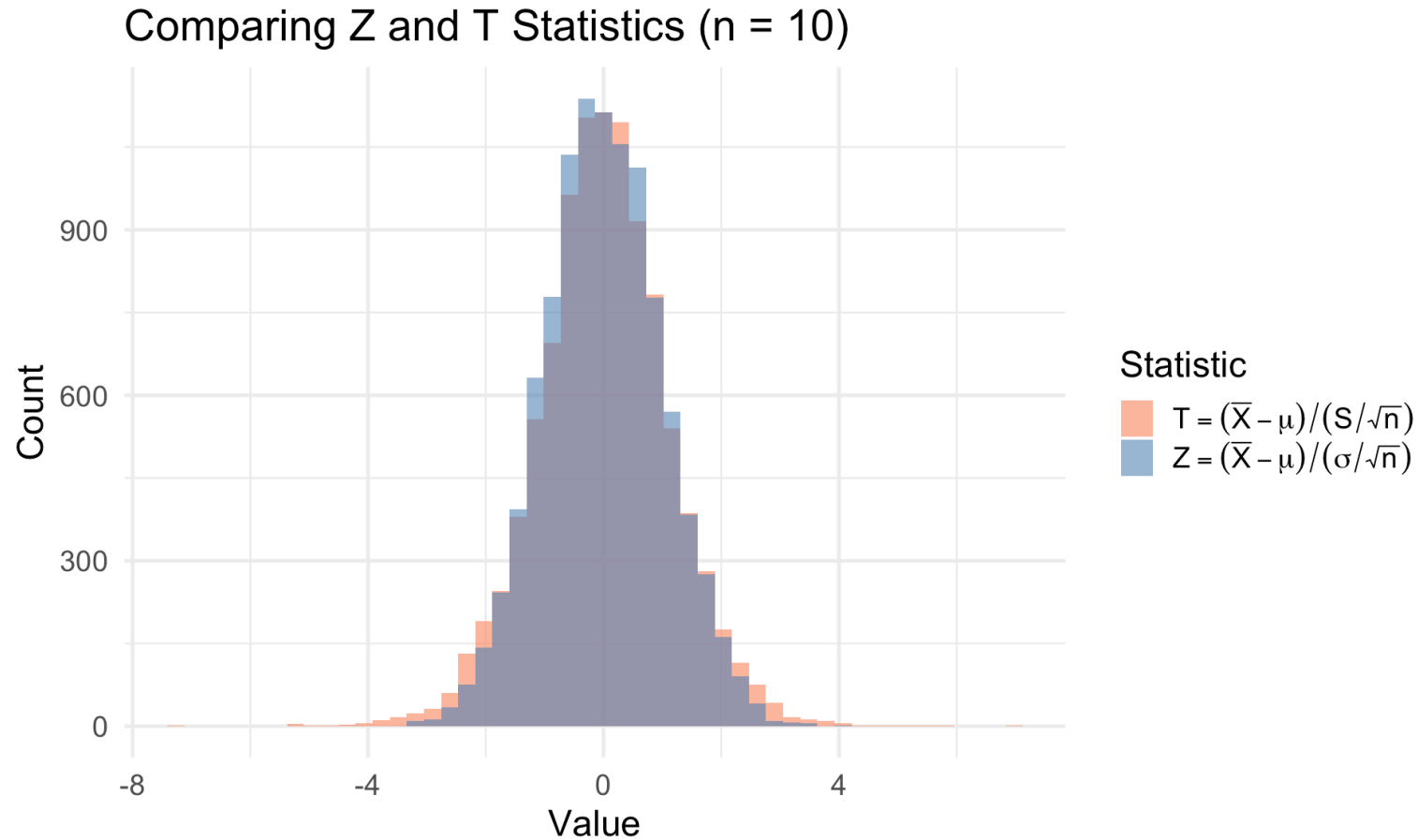
$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

becomes

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim ???$$

The Source of Extra Variability

► Code

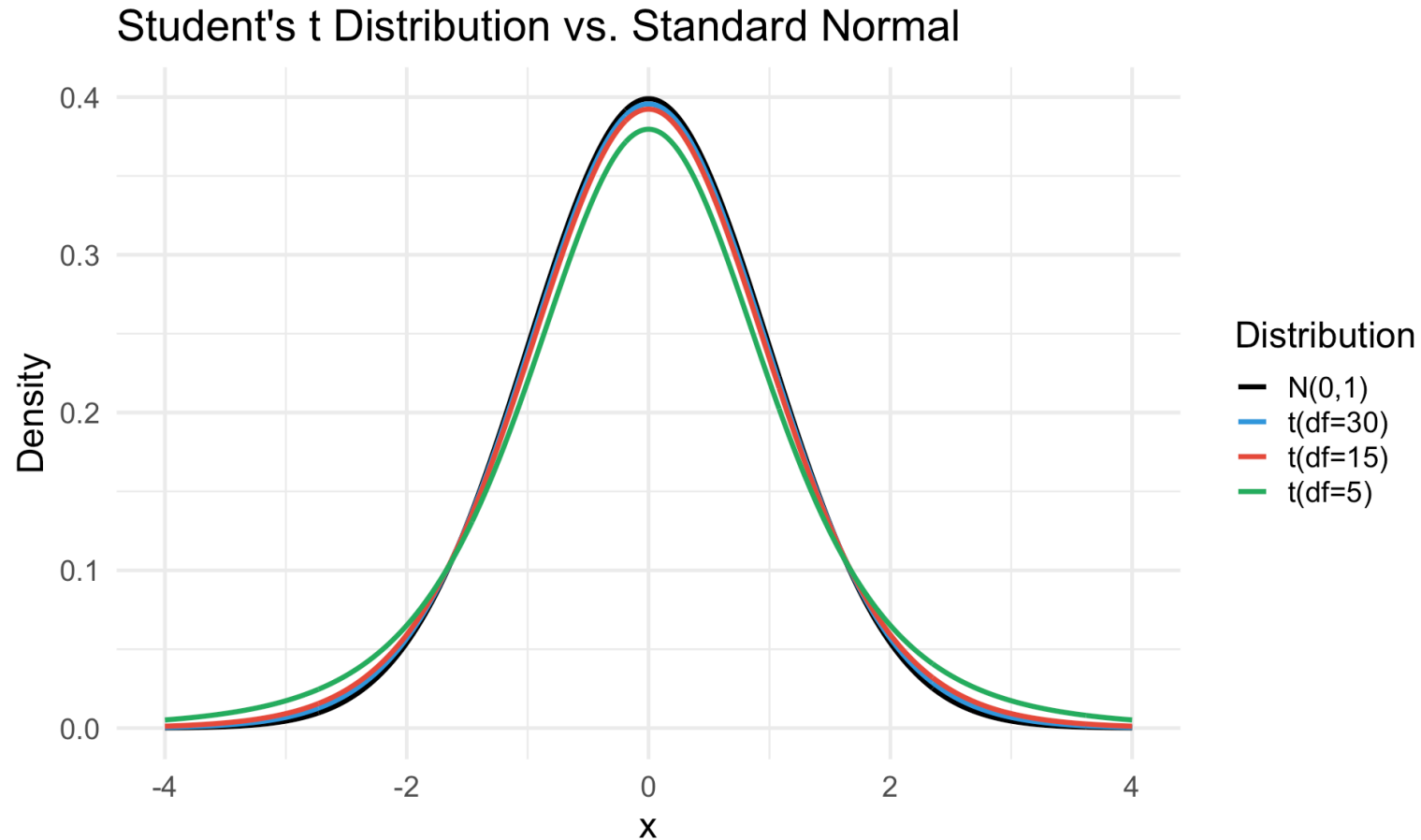


Key insight: T has more variability because both $\bar{X}$ and S are random!



The t Distribution

► Code



The t distribution has **heavier tails** than the normal — accounting for extra uncertainty from estimating σ .

Student's t Distribution

! Gosset's Theorem

If $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, then:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

where t_{n-1} denotes the t distribution with $n - 1$ **degrees of freedom**.

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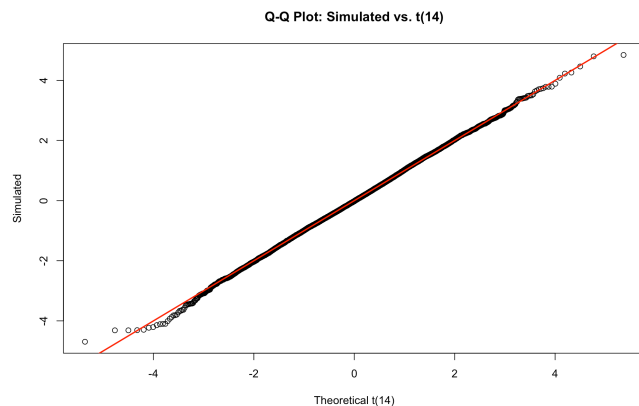
where t_{n-1} denotes the t distribution with $n - 1$ degrees of freedom.

Key properties of t_ν :

- Symmetric around 0
- Heavier tails than $N(0, 1)$
- As $\nu \rightarrow \infty$, $t_\nu \rightarrow N(0, 1)$

Simulating the t Distribution (Chihara 7.1.2)

```
1 # Verify that  $(\bar{X} - \mu)/(S/\sqrt{n})$  follows t distribution
2 set.seed(551)
3 n <- 15
4 t_stats <- replicate(10000, {
5   x <- rnorm(n, mean = 25, sd = 7)
6   (mean(x) - 25) / (sd(x) / sqrt(n))
7 })
8
9 # Compare to theoretical t distribution
10 qqplot(qt(ppoints(10000), df = n - 1), t_stats,
11        main = "Q-Q Plot: Simulated vs. t(14)",
12        xlab = "Theoretical t(14)", ylab = "Simulated")
13 abline(0, 1, col = "red", lwd = 2)
```



Why $(n - 1)$ Degrees of Freedom?

The sample variance involves:

$$S^2 = \frac{1}{n - 1} \sum_{i=1}^n (X_i - \bar{X})^2$$

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Intuition: We “use up” one degree of freedom by estimating μ with $\bar{X}$.

The One-Sample t Interval

! t Confidence Interval for Normal Mean (σ Unknown)

If $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, a $100(1 - \alpha)\%$ CI for μ is:

$$\bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}$$

where $t_{\alpha/2, n-1}$ is the critical value with $P(T > t_{\alpha/2, n-1}) = \alpha/2$ for $T \sim t_{n-1}$.

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The quantity $s/\sqrt{n}$ is called the **standard error of the mean** (SEM or SE).

t Critical Values

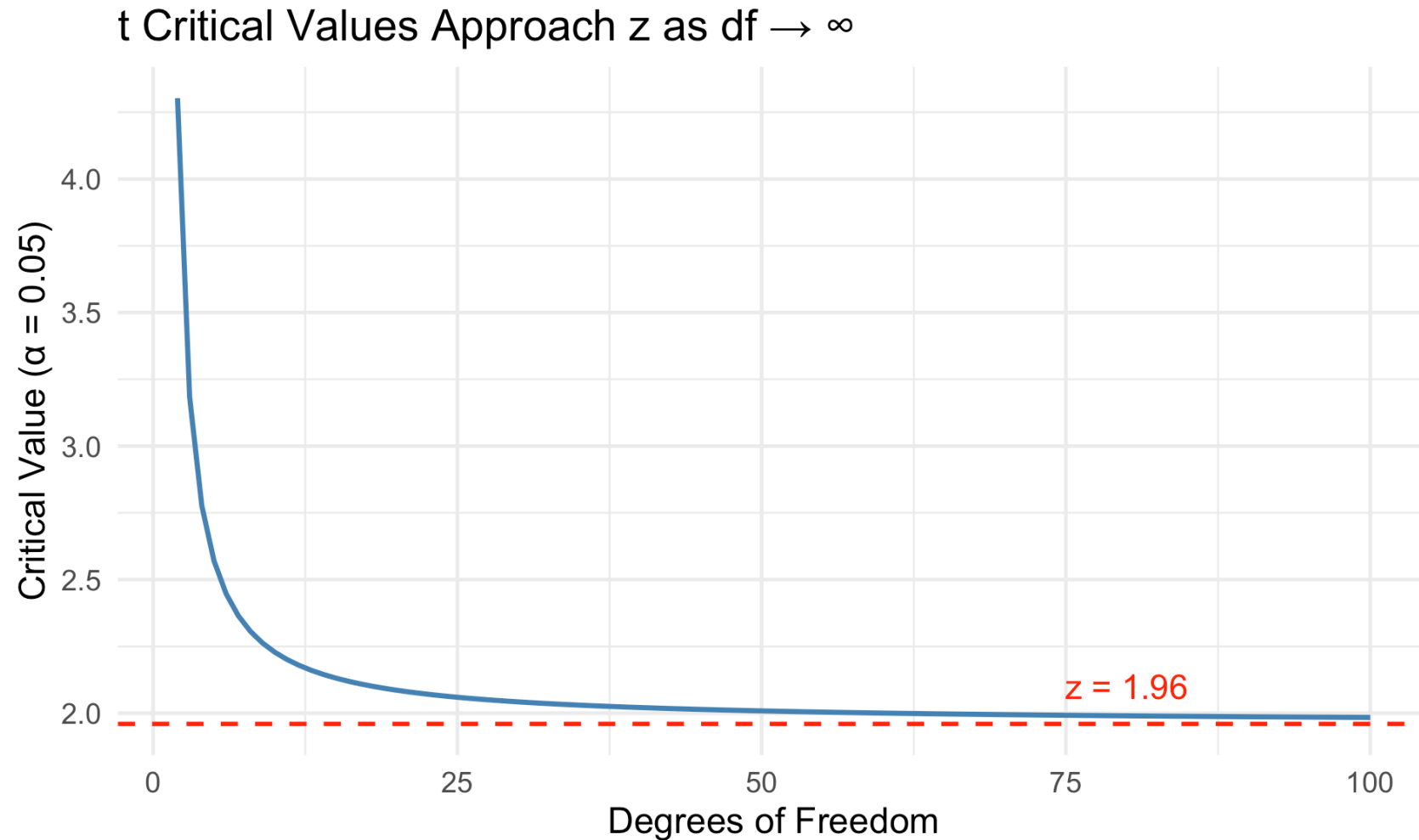
```
1 # Compare z and t critical values for 95% CI
2 z_crit <- qnorm(0.975)
3
4 df_values <- c(5, 10, 20, 30, 50, 100, Inf)
5 t_crits <- qt(0.975, df = df_values)
6
7 tibble(
8   df = c("5", "10", "20", "30", "50", "100", "∞ (z)"),
9   t_critical = round(c(head(t_crits, -1), z_crit), 3)
10 )
```

```
# A tibble: 7 × 2
  df      t_critical
<chr>      <dbl>
1 5          2.57
2 10         2.23
3 20         2.09
4 30         2.04
5 50         2.01
6 100        1.98
7 ∞ (z)      1.96
```

Key insight: For small samples, t critical values are substantially larger than z critical values.

Visualizing z vs. t Critical Values

► Code



Example: Porsche Odometer Readings (Devore Example 8.8)

A sample of $n = 16$ used Porsche Boxsters has odometer readings (miles):

```
1 odometer <- c(80000, 30100, 97500, 58551, 73787, 51800, 69267, 44530,  
2               42192, 104920, 41442, 27418, 43436, 77219, 5991, 14362)  
3  
4 n <- length(odometer)  
5 xbar <- mean(odometer)  
6 s <- sd(odometer)  
7  
8 cat("Sample size:", n, "\n")
```

Sample size: 16

```
1 cat("Sample mean:", round(xbar, 1), "miles\n")
```

Sample mean: 53907.2 miles

```
1 cat("Sample SD:", round(s, 1), "miles\n")
```

Sample SD: 28287.2 miles

Example: Porsche (continued)

95% Confidence Interval:

```
1 alpha <- 0.05
2 t_crit <- qt(1 - alpha/2, df = n - 1)
3
4 cat("t critical value (df=15):", round(t_crit, 3), "\n")
```

t critical value (df=15): 2.131

```
1 se <- s / sqrt(n)
2 margin <- t_crit * se
3
4 cat("Standard error:", round(se, 1), "\n")
```

Standard error: 7071.8

```
1 cat("Margin of error:", round(margin, 1), "\n\n")
```

Margin of error: 15073.2

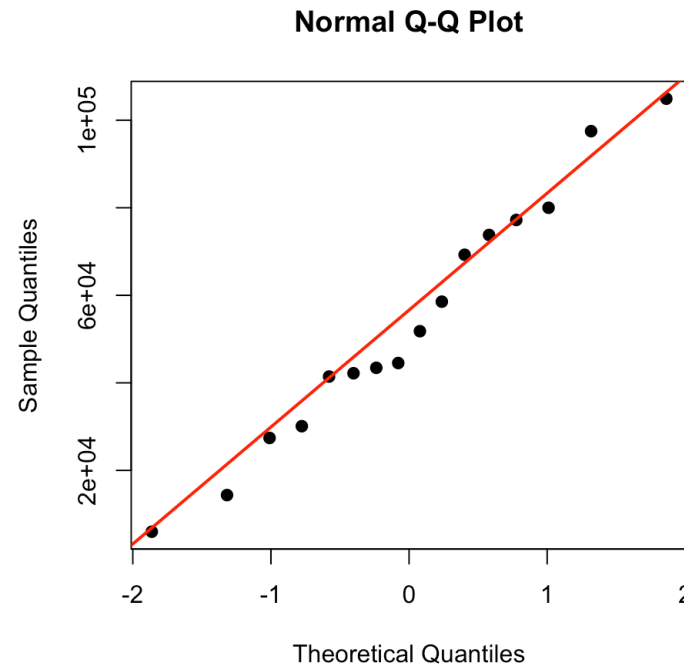
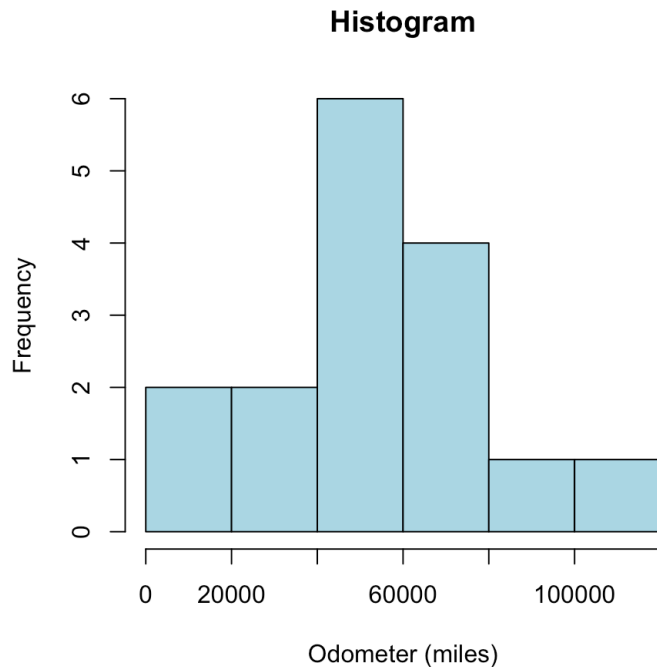
```
1 cat("95% CI: (", round(xbar - margin, 1), ",", round(xbar + margin, 1), ")")
```

95% CI: (38834 , 68980.4)

Checking the Normality Assumption

The t interval assumes normally distributed data. **Always check!**

```
1 par(mfrow = c(1, 2))
2 hist(odometer, main = "Histogram", xlab = "Odometer (miles)", col = "lightblue")
3 qqnorm(odometer, main = "Normal Q-Q Plot", pch = 19)
4 qqline(odometer, col = "red", lwd = 2)
```



The Q-Q plot looks reasonably linear — normality assumption is plausible.



Medical Example: Systolic Blood Pressure

A clinical study measures systolic blood pressure in $n = 25$ patients after a new treatment:

```
1 # Simulated BP data (mmHg)
2 bp <- c(118, 124, 131, 127, 122, 119, 128, 135, 121, 126,
3         130, 124, 129, 133, 120, 125, 127, 132, 123, 128,
4         126, 131, 122, 129, 125)
5
6 n <- length(bp)
7 xbar <- mean(bp)
8 s <- sd(bp)
9
10 # 95% CI
11 t_crit <- qt(0.975, df = n - 1)
12 se <- s / sqrt(n)
13 ci <- c(xbar - t_crit * se, xbar + t_crit * se)
14
15 cat("Sample mean:", round(xbar, 1), "mmHg\n")
```

Sample mean: 126.2 mmHg

```
1 cat("Sample SD:", round(s, 1), "mmHg\n")
```

Sample SD: 4.5 mmHg

```
1 cat("95% CI: (", round(ci[1], 1), ",", round(ci[2], 1), ") mmHg")
```

95% CI: (124.3 , 128.1) mmHg



Using `t.test()` in R

```
1 # The easy way to compute t confidence intervals
2 t.test(bp, conf.level = 0.95)
```

One Sample t-test

```
data:  bp
t = 140.51, df = 24, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 124.3463 128.0537
sample estimates:
mean of x
 126.2
```

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 124.3463 128.0537
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mean of x
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```

Note: R's `t.test()` also performs a hypothesis test (which we'll cover later).

Using broom to Tidy t.test() Output

The `broom` package provides a convenient way to extract results as a tibble:

```
1 # Tidy output with broom
2 library(broom)
3
4 t.test(bp, conf.level = 0.95) |>
5   tidy() |>
6   select(estimate, conf.low, conf.high, statistic, p.value)
```

A tibble: 1 × 5

	estimate	conf.low	conf.high	statistic	p.value
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	126.	124.	128.	141.	1.65e-36

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Advantages of broom:

- Results as a tibble (easy to manipulate)
- Consistent interface across many statistical functions
- Great for automated reporting and combining results

Medical Example: Stress and Cortisol (Devore 8.37)

Medical students' serum cortisol levels were measured at the beginning and end of semester. For $n = 41$ students, the mean difference (end - beginning) was $\bar{d} = 2.08$ with $s_d = 7.88$.

Find a 95% lower confidence bound for the population mean change:

```
1 n <- 41
2 dbar <- 2.08
3 s <- 7.88
4
5 t_alpha <- qt(0.95, df = n - 1) # One-sided!
6 se <- s / sqrt(n)
7
8 lower_bound <- dbar - t_alpha * se
9
10 cat("95% lower confidence bound:", round(lower_bound, 2), "\n")
```

95% lower confidence bound: 0.01

```
1 cat("Is the mean change necessarily positive? ",
2     ifelse(lower_bound > 0, "Yes", "No"))
```

Is the mean change necessarily positive? Yes

When Can We Trust the t-Interval?

Guidelines for t-Interval Validity

1. **Large samples ($n \geq 30$ -40):** t-interval is robust to non-normality (by CLT)
2. **Moderate samples (15-30):** Check for severe skewness or outliers
3. **Small samples ($n < 15$):** Normality is crucial; use Q-Q plots
4. **Outliers:** Can severely affect $\bar{x}$ and s ; consider alternatives

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Alternatives for non-normal data:

- Bootstrap confidence intervals (Lesson 9)
- Non-parametric methods (median-based)
- Transformations (log, square root)

Your Turn: Practice Problem

Exercise: A study of $n = 20$ hospital ICU stays yields $\bar{x} = 4.2$ days with $s = 1.8$ days.

1. Calculate a 95% CI for mean ICU stay
2. Calculate a 90% CI — how does it compare?
3. What assumptions are we making?

Your Turn: Practice Problem

Exercise: A study of $n = 20$ hospital ICU stays yields $\bar{x} = 4.2$ days with $s = 1.8$ days.

1. Calculate a 95% CI for mean ICU stay
2. Calculate a 90% CI — how does it compare?
3. What assumptions are we making?

```
1 n <- 20; xbar <- 4.2; s <- 1.8
2
3 # 95% CI
4 t95 <- qt(0.975, df = n - 1)
5 ci95 <- xbar + c(-1, 1) * t95 * s / sqrt(n)
6
7 # 90% CI
8 t90 <- qt(0.95, df = n - 1)
9 ci90 <- xbar + c(-1, 1) * t90 * s / sqrt(n)
10
11 cat("95% CI:", round(ci95, 2), "\n")
```

95% CI: 3.36 5.04

```
1 cat("90% CI:", round(ci90, 2))
```

90% CI: 3.5 4.9



Confidence Intervals for Proportions

The Setup

Situation: We observe X successes in n independent Bernoulli trials.

- $X \sim \text{Binomial}(n, p)$
- Point estimate: $\hat{p} = X/n$

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Examples in medical research:

- Proportion of patients responding to treatment
- Proportion experiencing adverse events
- Disease prevalence in a population
- Sensitivity/specificity of diagnostic tests

Deriving the MLE for Bernoulli (Review)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bernoulli}(p)$ where $P(X_i = 1) = p$.

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$$L(p) = \prod_{i=1}^n p^{x_i} (1 - p)^{1-x_i} = p^{\sum x_i} (1 - p)^{n - \sum x_i}$$

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Log-likelihood:

$$\ell(p) = \left(\sum x_i \right) \ln(p) + \left(n - \sum x_i \right) \ln(1 - p)$$

MLE for Bernoulli (continued)

Taking the derivative and setting to zero:

$$\frac{d\ell}{dp} = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1 - p} = 0$$

MLE for Bernoulli (continued)

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Solving for p :

$$\frac{\sum x_i}{p} = \frac{n - \sum x_i}{1 - p}$$

$$(1 - p) \sum x_i = p(n - \sum x_i)$$

$$\sum x_i = np$$

MLE for Bernoulli (continued)

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! MLE for Bernoulli Parameter

$$\hat{p}_{MLE} = \frac{\sum X_i}{n} = \bar{X} = \frac{X}{n}$$

Verifying $\hat{p}$ is a Maximum

Second derivative test: We need $\ell''(p) < 0$ at $\hat{p}$.

$$\ell'(p) = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1 - p}$$

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$$\ell''(p) = -\frac{\sum x_i}{p^2} - \frac{n - \sum x_i}{(1 - p)^2}$$

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$$\ell''(p) = -\frac{\sum x_i}{p^2} - \frac{n - \sum x_i}{(1 - p)^2}$$

Since $\sum x_i \geq 0$ and $n - \sum x_i \geq 0$, we have $\ell''(p) \leq 0$ for all $p \in (0, 1)$.

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When $0 < \hat{p} < 1$: $\ell''(\hat{p}) < 0 \checkmark \rightarrow$ confirms $\hat{p}$ is a **maximum**.

The General Wald Interval

! Wald Confidence Interval (General Definition)

For an MLE $\hat{\theta}$ with large n , the **Wald interval** is based on the asymptotic normality of MLEs:

$$\hat{\theta} \sim N\left(\theta, \frac{1}{I_n(\theta)}\right)$$

A $100(1 - \alpha)\%$ Wald CI is:

$$\hat{\theta} \pm z_{\alpha/2} \cdot \widehat{SE}(\hat{\theta})$$

where $\widehat{SE}$ is an estimate of the standard error, typically using $\hat{\theta}$ in place of θ .

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Key idea: Replace unknown parameters in the standard error with their MLEs.

Why the Wald Interval Works: Asymptotic Theory

From Lesson 4, we know the MLE has these asymptotic properties:

1. **Consistency:** $\hat{\theta} \xrightarrow{P} \theta$

2. **Asymptotic normality:** $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1/I(\theta))$

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For the Bernoulli case with $\hat{p} = \bar{X}$:

- $E(\hat{p}) = p$ and $\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$
- Fisher information: $I(p) = \frac{1}{p(1-p)}$, so $I_n(p) = \frac{n}{p(1-p)}$
- CRLB: $\frac{1}{I_n(p)} = \frac{p(1-p)}{n} = \text{Var}(\hat{p}) \checkmark$

The Wald Pivot

! Definition: Wald Pivot

The **Wald pivot** for a proportion is:

$$W = \frac{\hat{p} - p}{\widehat{SE}(\hat{p})} = \frac{\hat{p} - p}{\sqrt{\hat{p}(1 - \hat{p})/n}}$$

where we estimate the standard error by plugging in $\hat{p}$ for the unknown p .

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Large sample approximation: For large n ,

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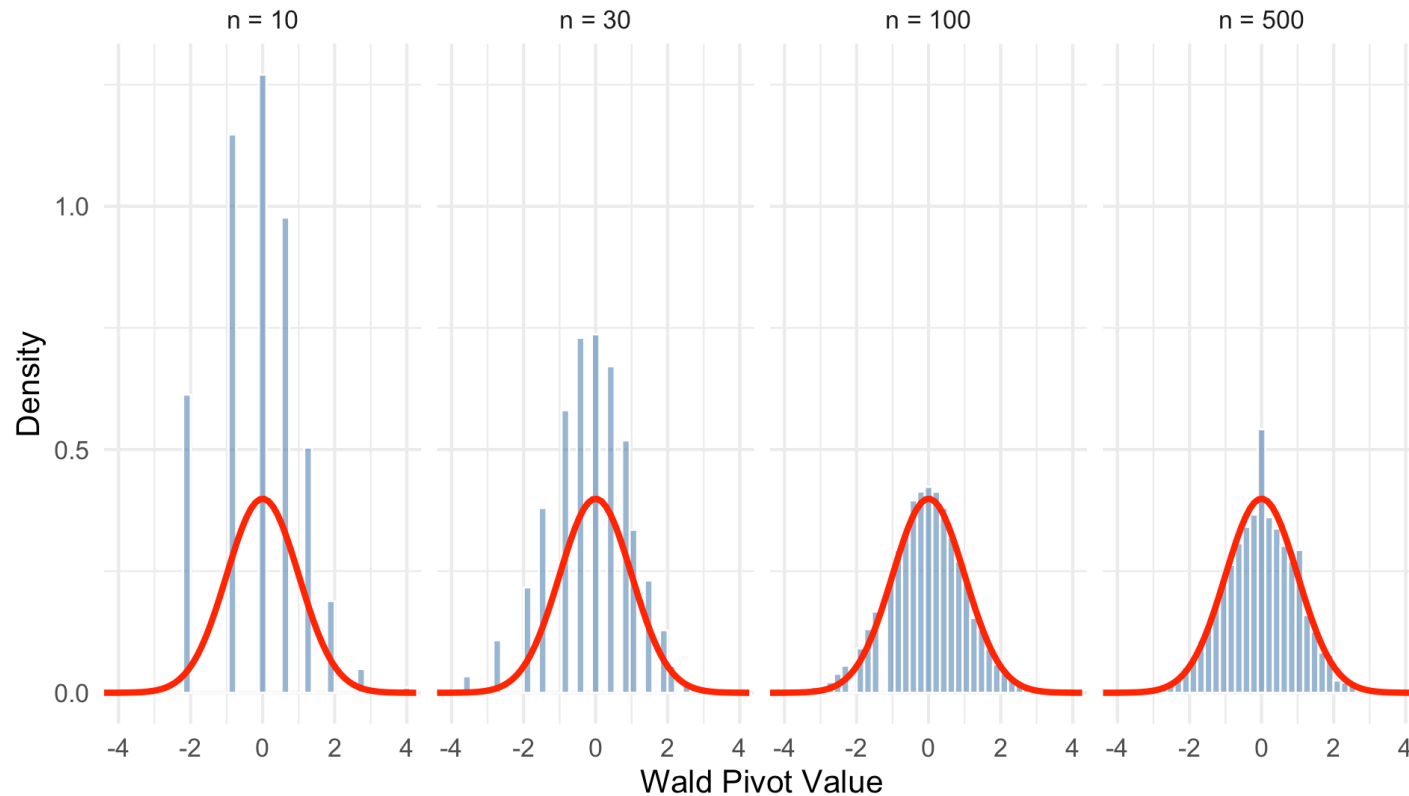
This approximation is what allows us to construct confidence intervals using $z_{\alpha/2}$.

Visualizing Convergence to $N(0,1)$: Density Comparison

► Code

Wald Pivot Distribution Converges to $N(0,1)$

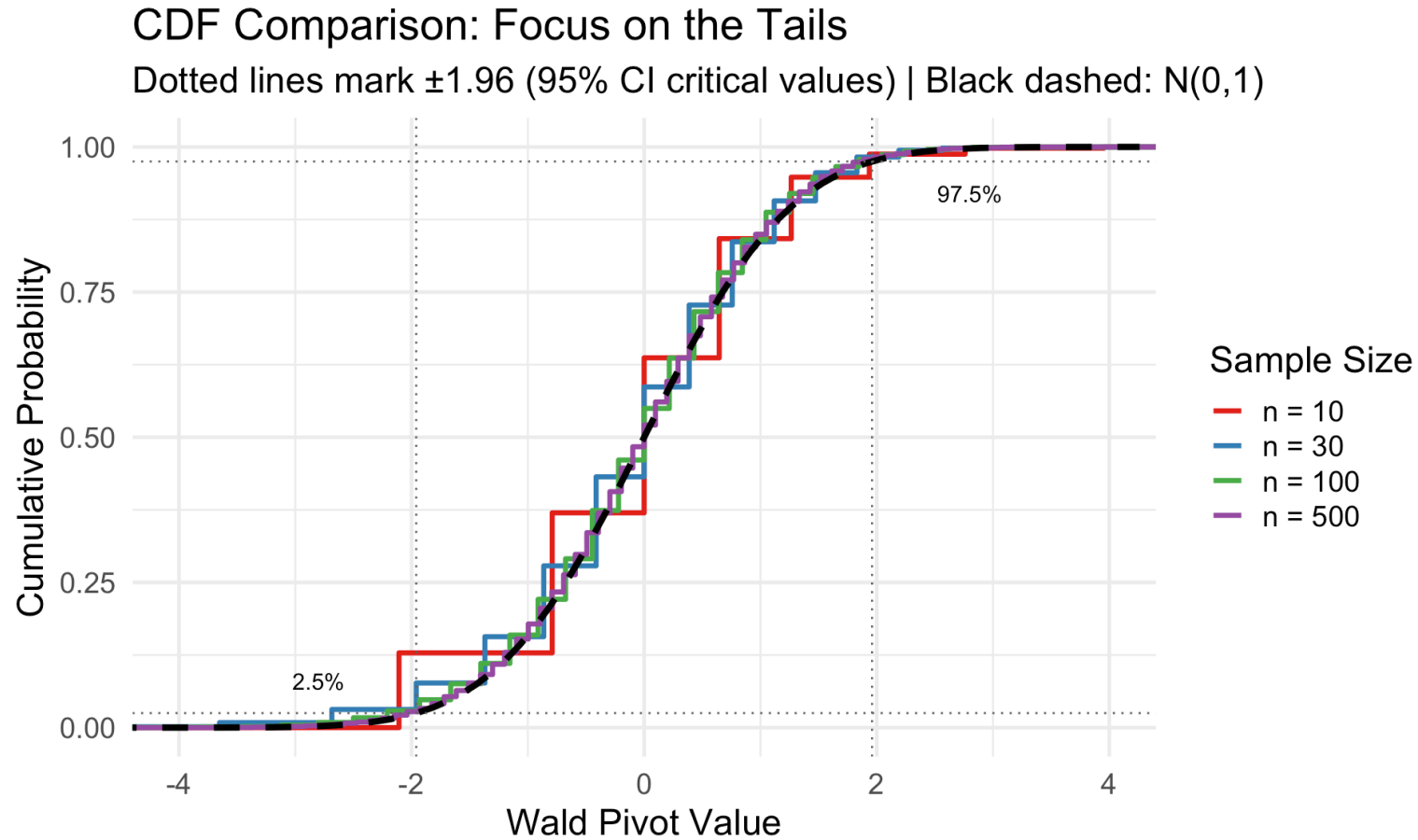
Blue histogram: 10,000 simulated pivots | Red curve: $N(0,1)$ density | True $p = 0.3$



As n increases, the histogram of simulated Wald pivots matches the red $N(0,1)$ curve more closely.

Convergence in the Tails (What Matters for CIs)

► Code

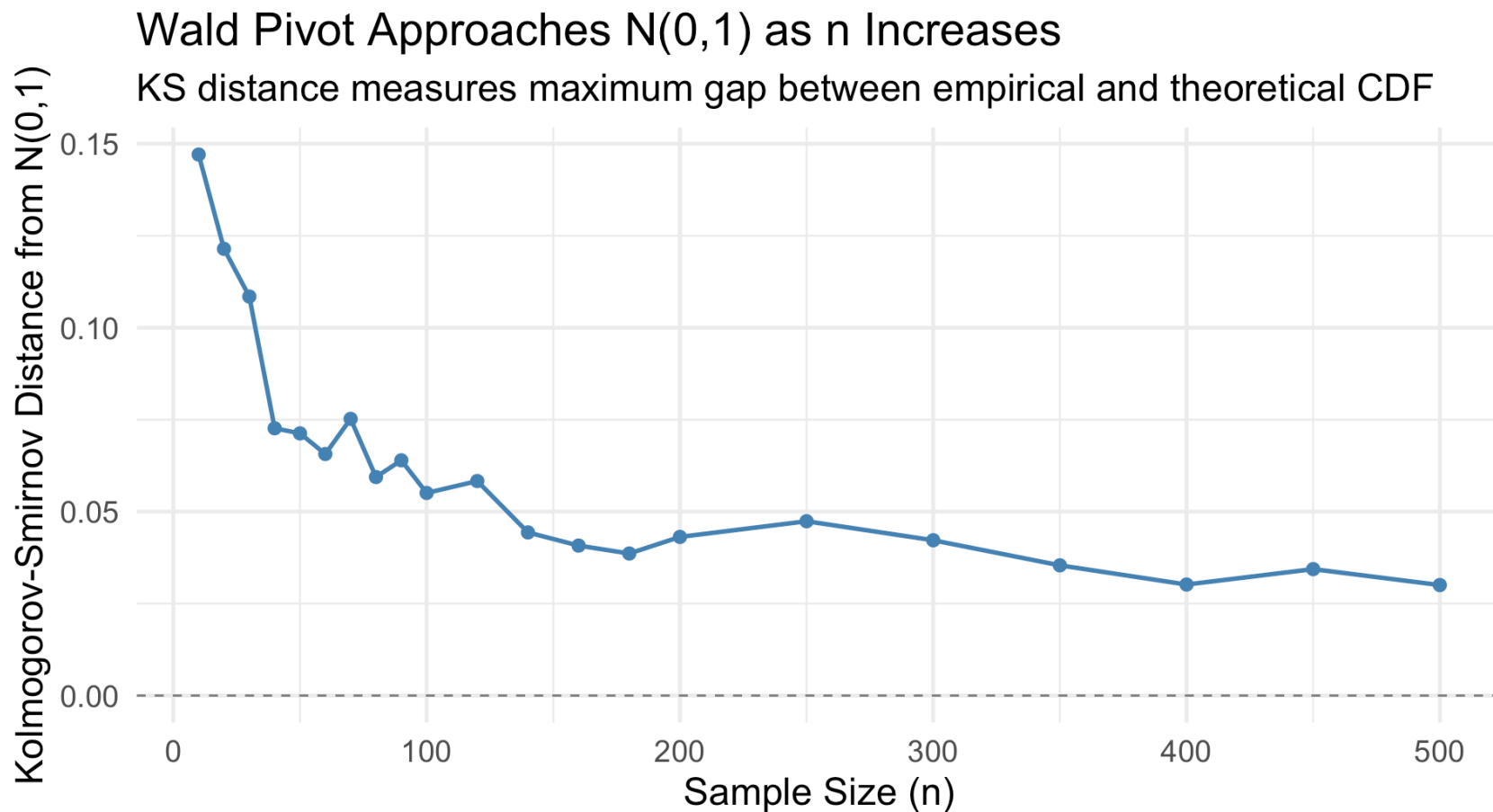


The tails (where we find critical values for CIs) converge more slowly than the center.



Quantifying Convergence: Kolmogorov-Smirnov Distance

► Code

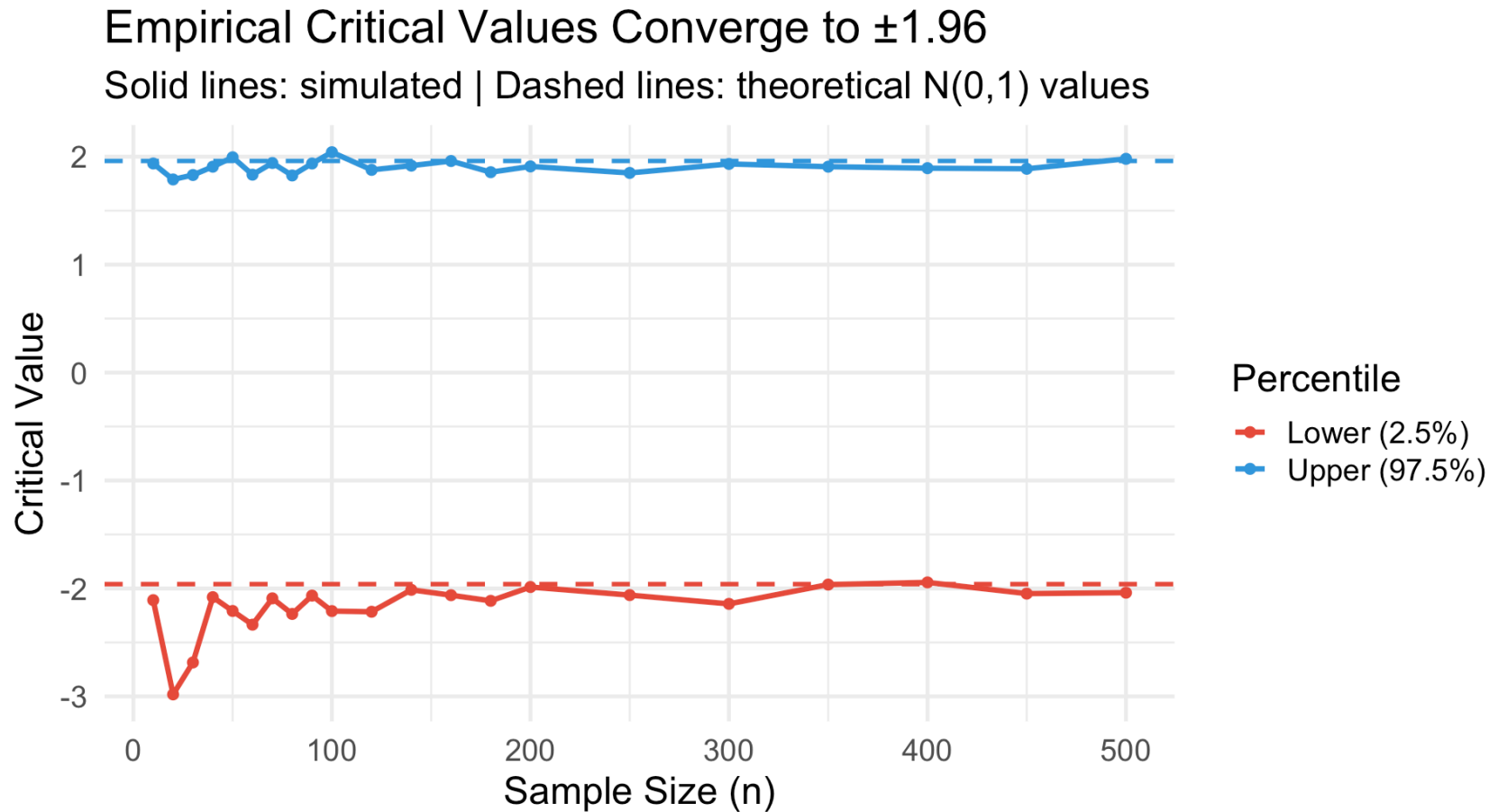


The KS distance (maximum difference between ECDFs) shrinks toward 0, confirming convergence in distribution.



Convergence: Critical Values for 95% CI

► Code



For 95% CIs, we need the 2.5th and 97.5th percentiles to equal ± 1.96 . This happens as $n \rightarrow \infty$.



The Exact Pivot for Proportions

If we knew p , we could use this **exact pivot**:

$$Z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \xrightarrow{d} N(0, 1)$$

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Problem: The standard error $\sqrt{p(1-p)/n}$ depends on the unknown p !

The Exact Pivot for Proportions

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$$Z = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \xrightarrow{d} N(0, 1)$$

Problem: The standard error $\sqrt{p(1-p)/n}$ depends on the unknown p !

Wald solution: Replace p with $\hat{p}$ in the standard error:

$$Z^* = \frac{\hat{p} - p}{\sqrt{\hat{p}(1-\hat{p})/n}}$$

But is Z^* still approximately $N(0, 1)$?

Why Plugging In $\hat{p}$ Works: Slutsky's Theorem

We can rewrite Z^* as:

$$Z^* = \frac{\hat{p} - p}{\sqrt{\hat{p}(1 - \hat{p})/n}} = \underbrace{\frac{\hat{p} - p}{\sqrt{p(1 - p)/n}}}_{\stackrel{d}{\rightarrow} N(0,1)} \cdot \underbrace{\frac{\sqrt{p(1 - p)}}{\sqrt{\hat{p}(1 - \hat{p})}}}_{\stackrel{P}{\rightarrow} 1}$$

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Slutsky's Theorem: If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{P} c$ (a constant), then:

$$X_n \cdot Y_n \xrightarrow{d} c \cdot X$$

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Since $\hat{p} \xrightarrow{P} p$ (consistency), we have:

$$\frac{\sqrt{p(1 - p)}}{\sqrt{\hat{p}(1 - \hat{p})}} \xrightarrow{P} 1$$

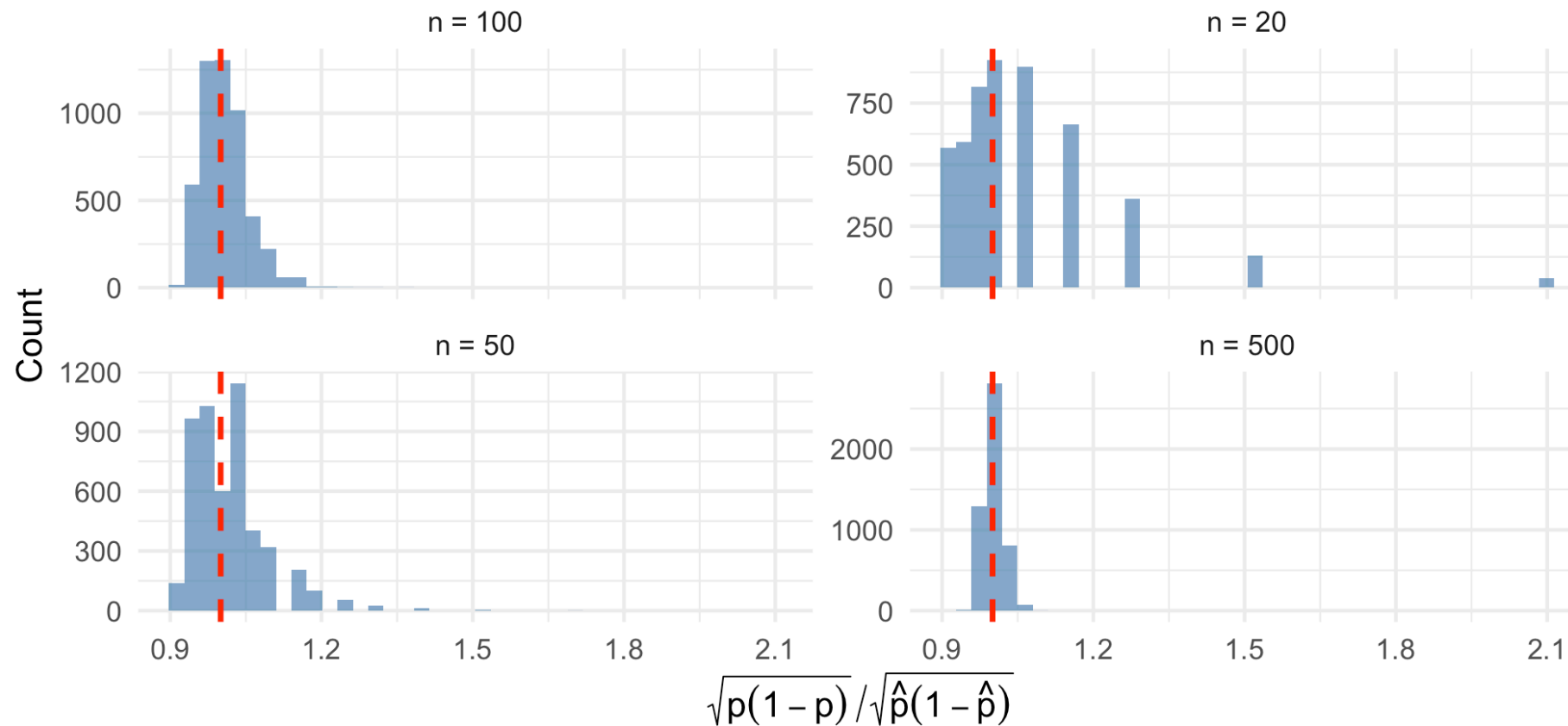
Therefore: $Z^* \xrightarrow{d} N(0, 1)$

Visualizing the Consistency Argument

► Code

Ratio of True SE to Estimated SE Concentrates at 1

True $p = 0.3$



The Wald Interval for Proportions

Using $Z^* = \frac{\hat{p} - p}{\sqrt{\hat{p}(1-\hat{p})/n}} \approx N(0, 1)$ as our pivot:

$$P \left(-z_{\alpha/2} < \frac{\hat{p} - p}{\sqrt{\hat{p}(1-\hat{p})/n}} < z_{\alpha/2} \right) \approx 1 - \alpha$$

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Solving for p :

! Wald Confidence Interval for a Proportion

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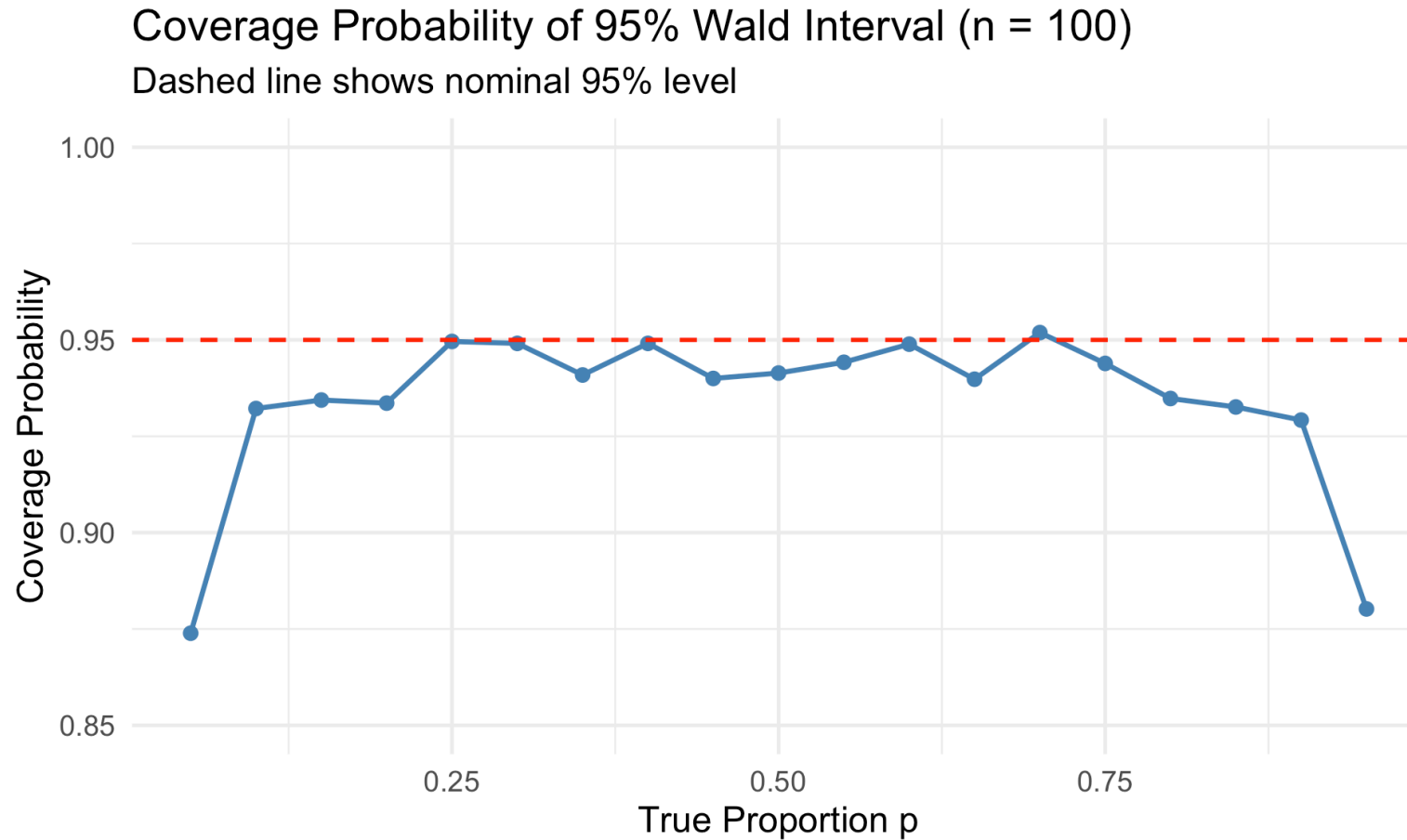
$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

! Problems with the Wald Interval

- Poor coverage probability for p near 0 or 1
- Can produce bounds outside $[0, 1]$
- Not recommended for small samples

Coverage Problems with Wald

► Code



The Wald interval often has coverage **below** the nominal level!

The Score Interval: A Better Approach

! The Score Test Statistic (General Definition)

The **score function** is the derivative of the log-likelihood:

$$U(\theta) = \ell'(\theta) = \frac{d}{d\theta} \ln L(\theta)$$

The **score test statistic** is:

$$Z_{score} = \frac{U(\theta)}{\sqrt{I_n(\theta)}} \xrightarrow{d} N(0, 1)$$

where $I_n(\theta)$ is the Fisher information.

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where $I_n(\theta)$ is the Fisher information.

Key insight: The score statistic is evaluated at the **null/hypothesized** parameter value, not at the MLE!

Score Function for Bernoulli

For $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bernoulli}(p)$:

Log-likelihood: $\ell(p) = X \ln(p) + (n - X) \ln(1 - p)$ where $X = \sum X_i$

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Score function:

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Fisher information: $I_n(p) = \frac{n}{p(1-p)}$

Score statistic:

$$Z_{score} = \frac{U(p)}{\sqrt{I_n(p)}} = \frac{(X - np)/[p(1 - p)]}{\sqrt{n/[p(1 - p)]}} = \frac{X - np}{\sqrt{np(1 - p)}} = \frac{\hat{p} - p}{\sqrt{p(1 - p)/n}}$$

Score Interval: Keep p in the Denominator!

Notice that the score statistic for proportions is:

$$Z_{score} = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \approx N(0, 1)$$

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$$\left| \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \right| \leq z_{\alpha/2}$$

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Squaring both sides:

$$\frac{(\hat{p} - p)^2}{p(1-p)/n} \leq z_{\alpha/2}^2$$

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Squaring both sides:

$$\frac{(\hat{p} - p)^2}{p(1-p)/n} \leq z_{\alpha/2}^2$$

This is a **quadratic equation** in p !

Solving the Quadratic for the Score Interval

Rearranging $(\hat{p} - p)^2 \leq z^2 \cdot p(1 - p)/n$:

$$n(\hat{p} - p)^2 \leq z^2 p(1 - p)$$

$$n\hat{p}^2 - 2np\hat{p} + np^2 \leq z^2 p - z^2 p^2$$

$$(n + z^2)p^2 - (2n\hat{p} + z^2)p + n\hat{p}^2 \leq 0$$

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$$(n + z^2)p^2 - (2n\hat{p} + z^2)p + n\hat{p}^2 \leq 0$$

Solving with the quadratic formula gives the **Score (Wilson) interval**:

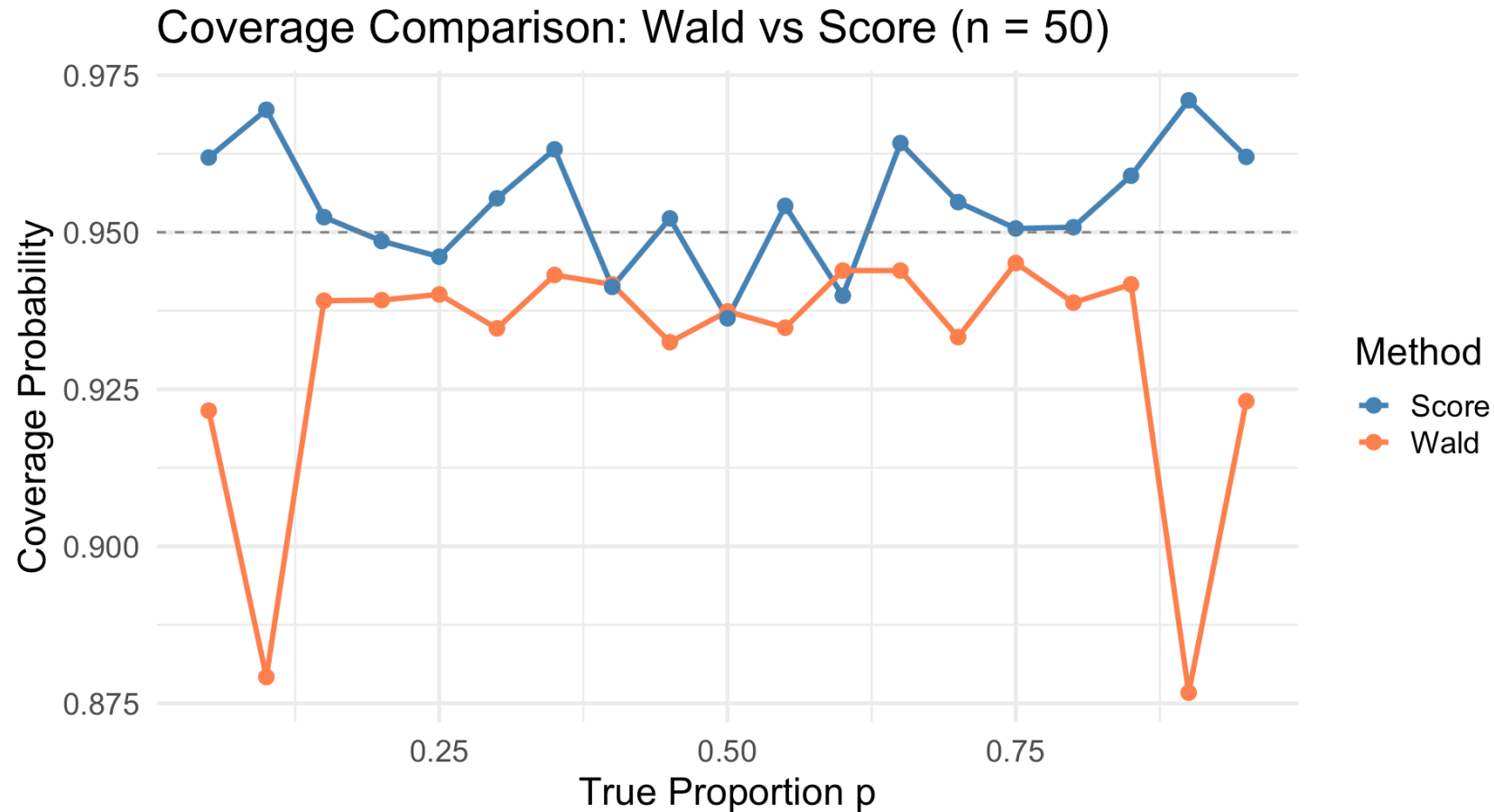
! Score Confidence Interval for a Proportion

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$$

where $z = z_{\alpha/2}$.

Why Score Intervals Are Better

► Code



Score interval maintains coverage closer to nominal level, especially for extreme p .

The Agresti-Coull Interval

A simpler approximation that works well in practice:

💡 Agresti-Coull (“Add Two Successes and Two Failures”)

Let $\tilde{n} = n + 4$ and $\tilde{p} = \frac{X+2}{n+4}$

Then use:

$$\tilde{p} \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}}$$

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Why it works: Adding 2 successes and 2 failures “shrinks” extreme estimates toward 0.5, improving coverage.

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Why it works: Adding 2 successes and 2 failures “shrinks” extreme estimates toward 0.5, improving coverage.

Connection to Score: With $z_{0.025} \approx 2$, the Score interval is approximately Agresti-Coull!

Comparing Intervals: Clinical Trial Example

In a Phase II trial, 23 of 40 patients respond to treatment.

```
1 x <- 23; n <- 40; p_hat <- x/n
2 cat("Sample proportion:", p_hat, "\n\n")
```

Sample proportion: 0.575

```
1 # Wald interval
2 se_wald <- sqrt(p_hat * (1 - p_hat) / n)
3 wald_ci <- p_hat + c(-1, 1) * 1.96 * se_wald
4
5 # Agresti-Coull interval
6 n_tilde <- n + 4
7 p_tilde <- (x + 2) / n_tilde
8 se_ac <- sqrt(p_tilde * (1 - p_tilde) / n_tilde)
9 ac_ci <- p_tilde + c(-1, 1) * 1.96 * se_ac
10
11 cat("Wald 95% CI:      (", round(wald_ci[1], 3), ",", round(wald_ci[2], 3), ")\n")
```

Wald 95% CI: (0.422 , 0.728)

```
1 cat("Agresti-Coull 95% CI: (", round(ac_ci[1], 3), ",", round(ac_ci[2], 3), ")\n")
```

Agresti-Coull 95% CI: (0.422 , 0.715)

Using prop.test() for Score Interval

```
1 # Score (Wilson) interval in R
2 prop.test(x = 23, n = 40, conf.level = 0.95, correct = FALSE)
```

1-sample proportions test without continuity correction

```
data: 23 out of 40, null probability 0.5
X-squared = 0.9, df = 1, p-value = 0.3428
alternative hypothesis: true p is not equal to 0.5
95 percent confidence interval:
 0.4219507 0.7149061
sample estimates:
      p
0.575
```

Using broom with prop.test()

```
1 # Tidy output with broom
2 prop.test(x = 23, n = 40, conf.level = 0.95, correct = FALSE) |>
3   tidy() |>
4   select(estimate, conf.low, conf.high, statistic, p.value, method)
```

A tibble: 1 × 6

	estimate	conf.low	conf.high	statistic	p.value	method
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
1	0.575	0.422	0.715	0.9	0.343	1-sample proportions test witho...

Using broom with prop.test()

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Note: The `estimate` column shows $\hat{p}$, and `conf.low/conf.high` give the Score interval bounds.

Comparing All Three Methods

```
1 compare_proportion_cis <- function(x, n, conf = 0.95) {
2   p_hat <- x/n
3   z <- qnorm(1 - (1-conf)/2)
4
5   # Wald
6   se_w <- sqrt(p_hat * (1-p_hat) / n)
7   wald <- p_hat + c(-1, 1) * z * se_w
8
9   # Agresti-Coull
10  p_ac <- (x + 2) / (n + 4)
11  se_ac <- sqrt(p_ac * (1-p_ac) / (n + 4))
12  agresti <- p_ac + c(-1, 1) * z * se_ac
13
14  # Score (Wilson)
15  score <- prop.test(x, n, conf.level = conf, correct = FALSE)$conf.int
16
17  tibble(Method = c("Wald", "Agresti-Coull", "Score"),
18         Lower = c(wald[1], agresti[1], score[1]),
19         Upper = c(wald[2], agresti[2], score[2]))
20 }
21
22 compare_proportion_cis(23, 40)
```

Medical Example: Adverse Event Rate

In a safety study, 8 out of 200 patients experience a serious adverse event.

```
1 x <- 8; n <- 200
2
3 # Score interval
4 result <- prop.test(x, n, conf.level = 0.95, correct = FALSE)
5 cat("Point estimate:", x/n, "=", round(x/n * 100, 1), "%\n")
```

Point estimate: 0.04 = 4 %

```
1 cat("95% Score CI:", round(result$conf.int * 100, 2), "%")
```

95% Score CI: 2.04 7.69 %

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Point estimate: 0.04 = 4 %

```
1 cat("95% Score CI:", round(result$conf.int * 100, 2), "%")
```

95% Score CI: 2.04 7.69 %

Clinical interpretation: We are 95% confident that the true rate of serious adverse events is between 1.8% and 7.7%.

Extreme Proportions: Why Wald Fails

```
1 # What if only 2 of 50 patients have adverse events?
2 x <- 2; n <- 50; p_hat <- x/n
3
4 # Wald can give negative lower bound!
5 se_wald <- sqrt(p_hat * (1-p_hat) / n)
6 wald_ci <- p_hat + c(-1, 1) * 1.96 * se_wald
7
8 cat("Wald 95% CI:  (", round(wald_ci[1], 4), ",", round(wald_ci[2], 4), ")\n")
```

Wald 95% CI: (-0.0143 , 0.0943)

```
1 # Score interval stays in [0,1]
2 score_ci <- prop.test(x, n, conf.level = 0.95, correct = FALSE)$conf.int
3 cat("Score 95% CI:  (", round(score_ci[1], 4), ",", round(score_ci[2], 4), ")\n")
```

Score 95% CI: (0.011 , 0.1346)

The Wald interval gives a **negative** lower bound — impossible for a proportion!

Example: Super Bowl Viewing (Devore 8.15)

In a sample of 25 students, **all 25** said they watched the Super Bowl. What's a 95% CI for the true proportion?

```
1 x <- 25; n <- 25
2
3 # Wald gives silly result
4 p_hat <- x/n
5 se_wald <- sqrt(p_hat * (1-p_hat) / n) # This is 0!
6 cat("Wald 95% CI: (1, 1) – just the point estimate!\n\n")
```

Wald 95% CI: (1, 1) – just the point estimate!

```
1 # Score interval is much better
2 score_ci <- prop.test(x, n, conf.level = 0.95, correct = FALSE)$conf.int
3 cat("Score 95% CI: (", round(score_ci[1], 3), ",", round(score_ci[2], 3), ")")
```

Score 95% CI: (0.867 , 1)

Sample Size for Proportions

To achieve a margin of error b with confidence level $1 - \alpha$:

$$n = \frac{z_{\alpha/2}^2 \cdot p(1 - p)}{b^2}$$

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Sample Size for Proportions

To achieve a margin of error b with confidence level $1 - \alpha$:

$$n = \frac{z_{\alpha/2}^2 \cdot p(1 - p)}{b^2}$$

Problem: We don't know p before the study!

Conservative approach: Use $p = 0.5$ (maximizes $p(1 - p) = 0.25$):

$$n = \frac{z_{\alpha/2}^2 \cdot 0.25}{b^2}$$

Sample Size Example: Disease Prevalence

An epidemiologist wants to estimate disease prevalence with a margin of error $\leq 3\%$ with 95% confidence.

```
1 b <- 0.03 # margin of error
2 z <- qnorm(0.975)
3
4 # Conservative (p = 0.5)
5 n_conservative <- z^2 * 0.25 / b^2
6 cat("Conservative sample size:", ceiling(n_conservative), "\n")
```

Conservative sample size: 1068

```
1 # If we believe p ≈ 0.10 from prior studies
2 n_informed <- z^2 * 0.10 * 0.90 / b^2
3 cat("Informed sample size (p ≈ 0.10):", ceiling(n_informed))
```

Informed sample size (p ≈ 0.10): 385

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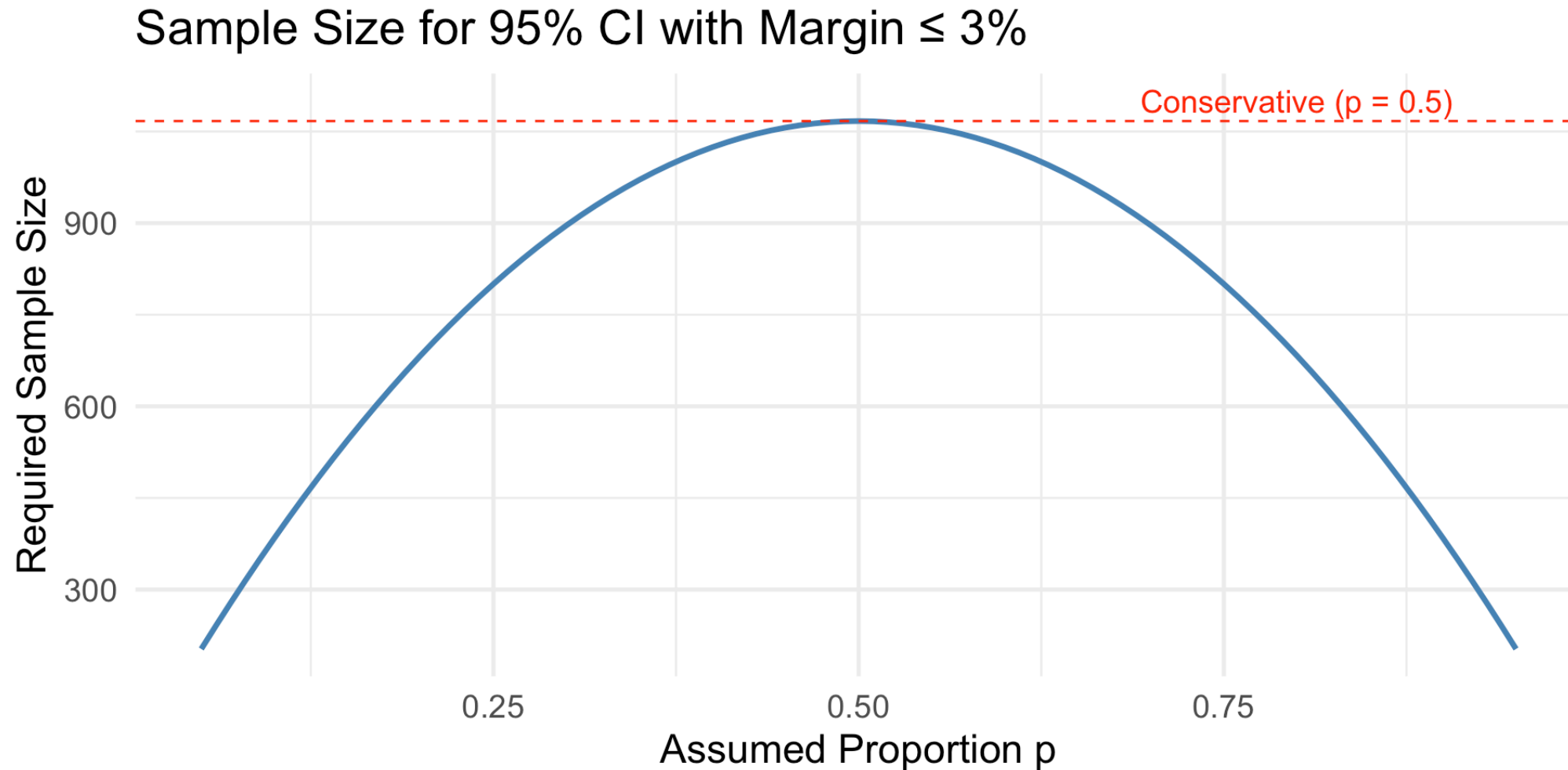
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3 cat("Informed sample size (p ≈ 0.10):", ceiling(n_informed))
```

Informed sample size (p ≈ 0.10): 385

Using prior information can **substantially** reduce required sample size!

Sample Size: Effect of Assumptions

► Code



One-Sided Bounds for Proportions

Upper confidence bound (e.g., maximum adverse event rate):

```
1 # 12 adverse events in 150 patients
2 x <- 12; n <- 150
3
4 # 95% upper bound using Score method
5 result <- prop.test(x, n, conf.level = 0.95,
6                     alternative = "less", correct = FALSE)
7 cat("Point estimate:", round(x/n * 100, 1), "%\n")
```

Point estimate: 8 %

```
1 cat("95% upper bound:", round(result$conf.int[2] * 100, 1), "%")
```

95% upper bound: 12.4 %

One-Sided Bounds for Proportions

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Point estimate: 8 %

```
1 cat("95% upper bound:", round(result$conf.int[2] * 100, 1), "%")
```

95% upper bound: 12.4 %

Interpretation: We are 95% confident that the true adverse event rate is **at most** 12.9%.

Practical Considerations

Summary: CI Methods Comparison

Method	When to Use	Notes
Z-interval	Normal data, σ known	Rarely applicable
t-interval	Normal data, σ unknown	Check normality
Wald (proportion)	Large n , p not extreme	Poor coverage
Score/Wilson	General	Best for proportions
Agresti-Coull	Quick calculation	Good approximation

CI Interpretation in Medical Research

Valid Interpretations

- “We are 95% confident that μ lies between L and U”
- “The interval (L, U) is a plausible range for μ ”
- “In repeated sampling, 95% of such intervals would contain μ ”

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- “The interval (L, U) is a plausible range for μ ”
- “In repeated sampling, 95% of such intervals would contain μ ”

Invalid Interpretations

- “There is a 95% probability that μ is in (L, U)” ✗
- “95% of the data falls between L and U” ✗
- “The true mean varies between L and U” ✗

Clinical Decision Making with CIs

Example: A drug reduces blood pressure by an estimated 12 mmHg with 95% CI (8, 16).

Clinical Decision Making with CIs

Example: A drug reduces blood pressure by an estimated 12 mmHg with 95% CI (8, 16).

Questions we can answer:

- Is the effect clinically meaningful? (If minimum important difference is 5 mmHg, yes!)
- Is zero a plausible value? (No — CI excludes 0)
- What's the worst-case scenario? (Lower bound suggests at least 8 mmHg effect)

Relationship to Hypothesis Testing

Preview of Coming Material

A 95% CI contains all parameter values that would **not** be rejected by a two-sided test at $\alpha = 0.05$.

- If CI excludes the null value $\rightarrow$ reject H_0
- If CI includes the null value $\rightarrow$ fail to reject H_0

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Preview of Coming Material

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- If CI excludes the null value $\rightarrow$ reject H_0
- If CI includes the null value $\rightarrow$ fail to reject H_0

Example: 95% CI for mean difference is (8, 16).

- Test $H_0 : \mu = 0 \rightarrow$ Reject (0 not in CI)
- Test $H_0 : \mu = 10 \rightarrow$ Fail to reject (10 in CI)

Summary and Looking Ahead

Lesson 8 Summary

Key Concepts:

1. **t-intervals** handle unknown σ : $\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$
2. **t distribution** has heavier tails than normal; approaches normal as $df \rightarrow \infty$
3. **Always check normality** for t-intervals (Q-Q plots, histograms)
4. **Wald intervals** based on asymptotic MLE theory; plug in $\hat{\theta}$ for SE
5. **Score intervals** keep the parameter in SE, solve quadratic; better coverage
6. **Agresti-Coull** (“add 2 and 2”) is a good simple approximation

Key Formulas

Parameter	CI Formula	Conditions
μ (σ known)	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	Normal data
μ (σ unknown)	$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$	Normal data
p (Wald)	$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	Large n
p (Score)	$\frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$	General
p (Agresti-Coull)	$\tilde{p} \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}}$	General

Looking Ahead

Next Lesson: Bootstrap Confidence Intervals

- What if normality doesn't hold?
- Bootstrap percentile intervals
- Bootstrap t intervals
- When to use bootstrap vs. classical methods

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Coming Soon:

- Hypothesis testing
- Two-sample inference
- Paired data analysis

References

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Questions?

Thank you!